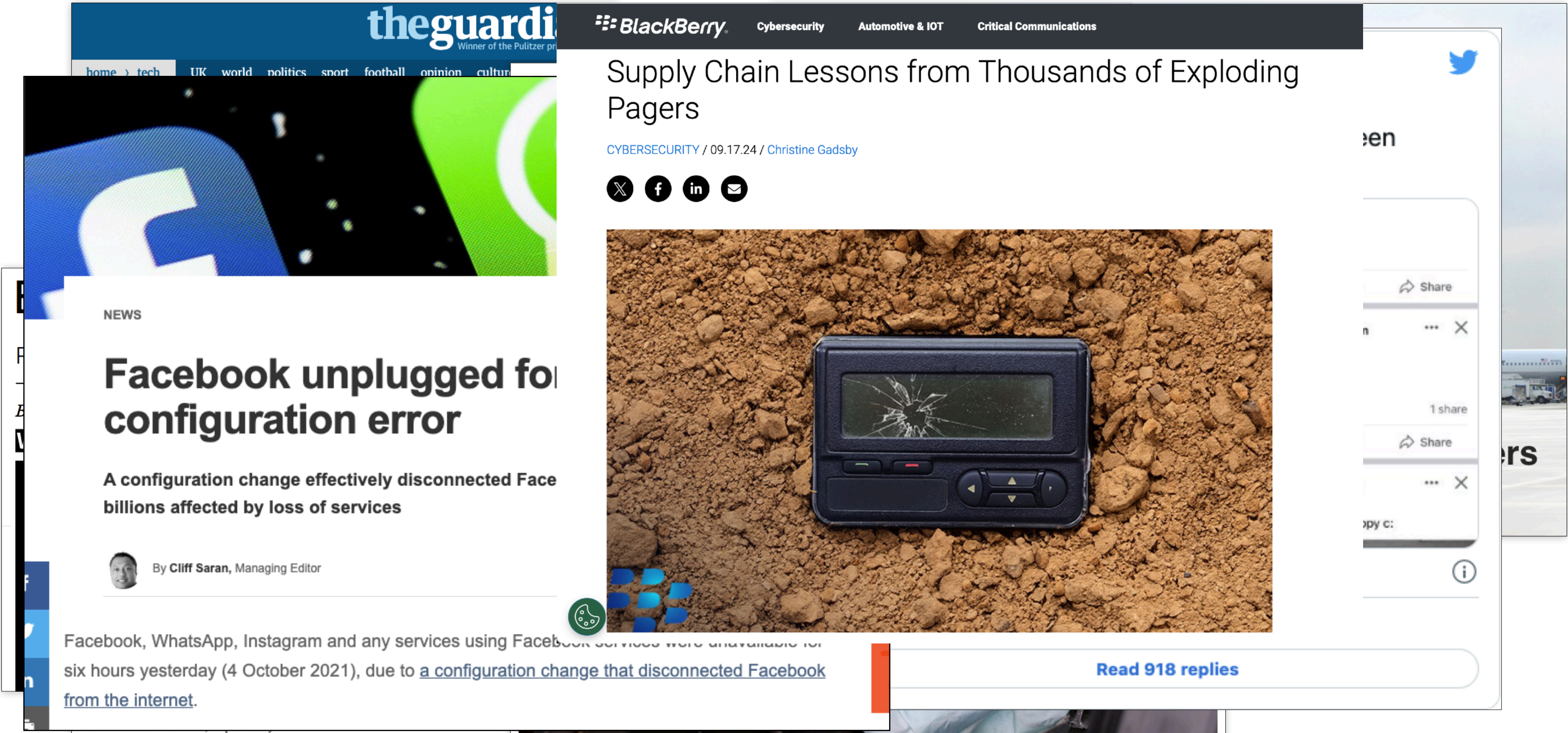
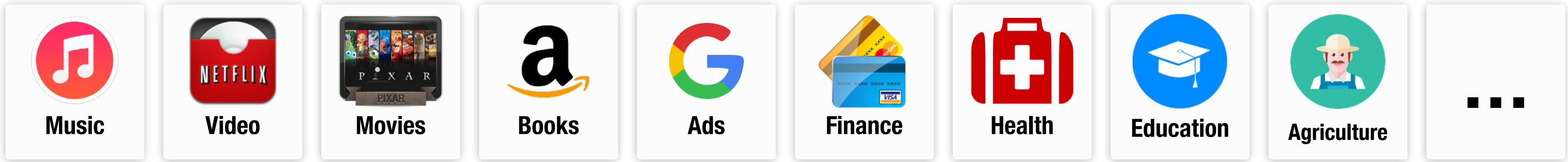


# **Software Dependability**

**Introduction to the course**

# Most businesses are being run on software with devastating risks due to software failures



# Such issues are not very far from us!

The image shows a screenshot of the INPS (Istituto Nazionale per lo Studio e la Cura delle Dipendenze) website. The website has a dark blue header with navigation links: 'Accedere ai servizi', 'Contatti', 'Trova la sede', and a language selector set to 'IT'. Below the header is a search bar with 'Cerca' and a login button 'Entra in MyINPS'. The main navigation menu includes 'Home', 'L'Istituto', 'Dati, ricerche e bilanci', 'INPS Comunica', and 'Prestazioni e servizi'. The main content area features a large banner with a virus icon and text about the 'Rapporto annuale 2020'. To the right, there is a sidebar with a list of services: 'Pensionati', 'Disoccupati, inoccupati e lavoratori sospesi', 'Persone con disabilità e invalidità', 'Lavoratori', and 'Lavoratori migranti'. Below this, there are sections for 'Finanziari' (Intermediari e consulenti, Medici e ASL) and 'Servizi più utilizzati'. Overlaid on the screenshot are five pink text boxes with white text, providing context about a data privacy breach. The first box mentions the COVID-19 lockdown and financial support. The second box states that people could request support through the INPS site. The third box notes that the website was accessed by thousands of people. The fourth box explains that the website crashed, but this was not the main problem. The fifth box describes a defect in the authentication mechanism that allowed access to other people's data, including bank account details. The final box states that this was considered the biggest violation of data privacy and security in Italian history.

Accedere ai servizi Contatti Trova la sede IT

Cerca Entra in MyINPS

Home L'Istituto Dati, ricerche e bilanci INPS Comunica Prestazioni e servizi

mercato del lavoro: pubblicato il Rapporto annuale 2020

Notizia 25/02/2021 Soci di società della pesca non cooperative: regime previdenziale

Pensionati Disoccupati, inoccupati e lavoratori sospesi Persone con disabilità e invalidità Lavoratori Lavoratori migranti

Finanziari Intermediari e consulenti Medici e ASL

Servizi più utilizzati

Ti potrebbero interessare

You might remember that during the covid19 lockdown, the Italian government approved financial support for the population.

People could request such financial support through the INPS site - the Italian national social security institution.

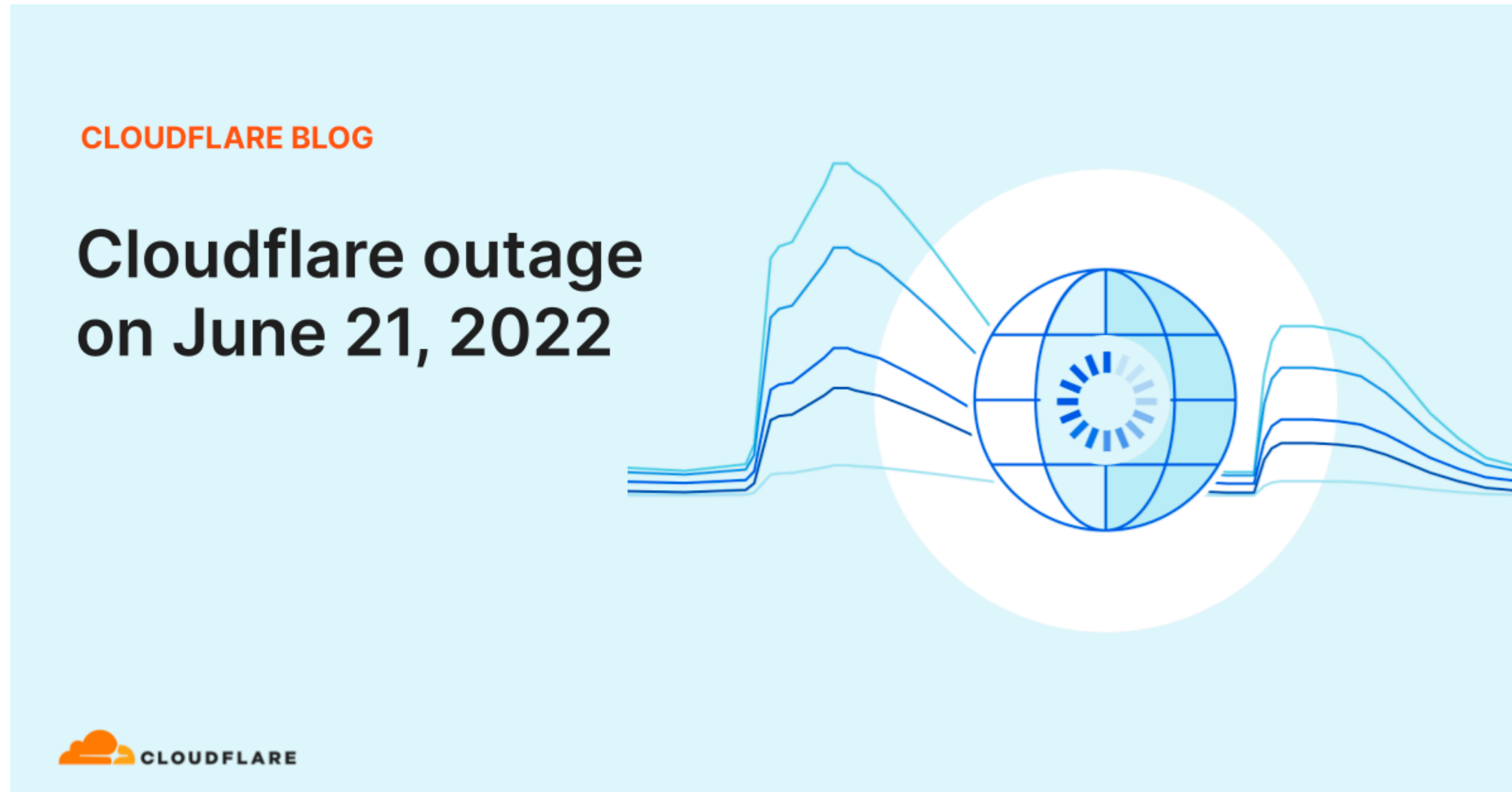
The website was accessed by thousands of people. Guess what?

The INPS website crashed, but this was NOT the main problem.

A defect in the authentication mechanism (apparently due to the lack of control for homonymous people) caused people to access other people's data - including bank account details!

According to someone, this has been the biggest violation of data privacy and security in Italian history.

# Outages related to Software Dependability are more and more frequent



<https://blog.cloudflare.com/cloudflare-outage-on-june-21-2022/>

**This is problematic, but there's something more...**

**2.41 trillion**

**in financial losses in 2022**

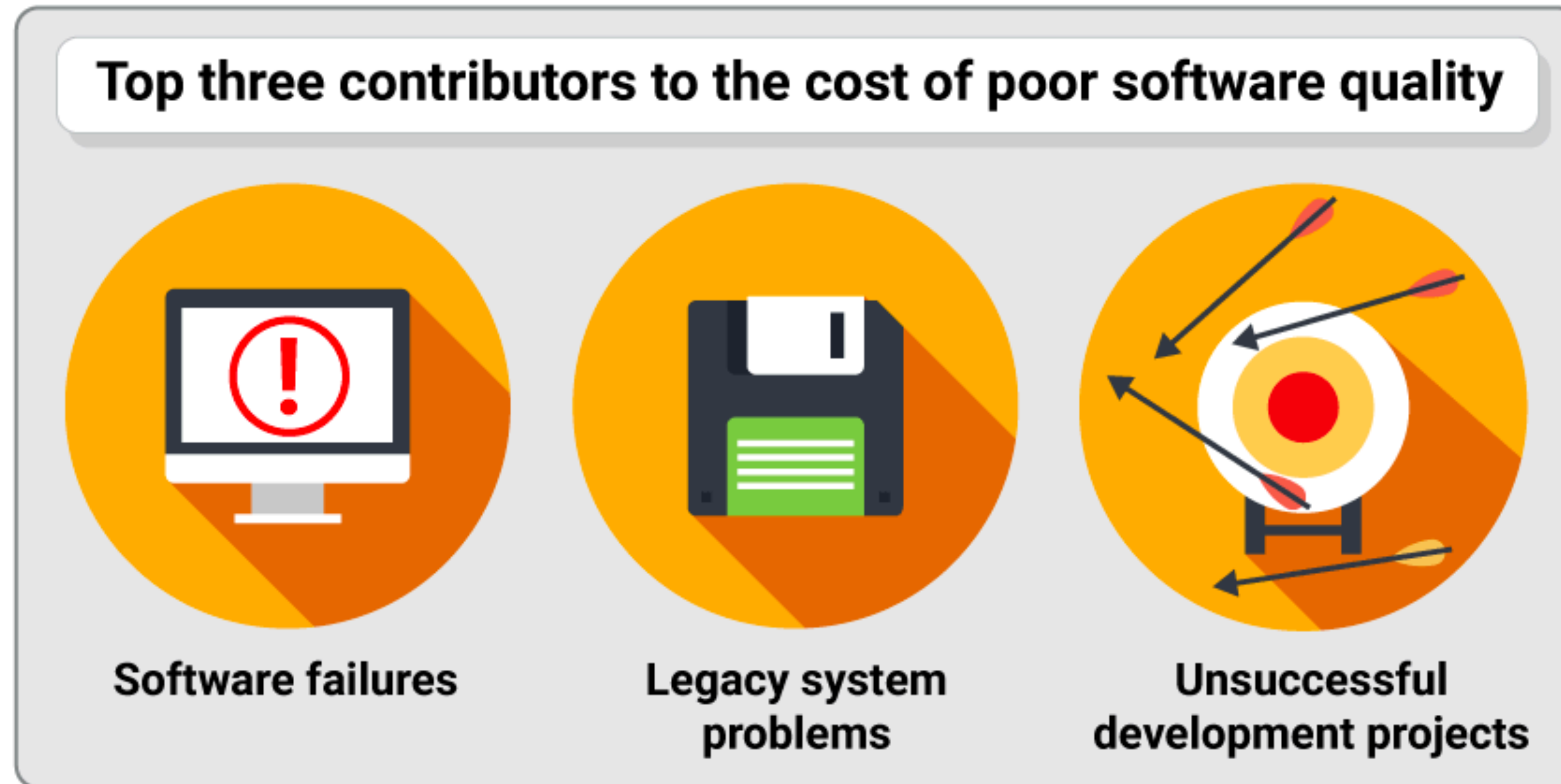
**7 trillion**

**in cybercrime losses in 2022 due to the rising number of software vulnerabilities**

**Synopsys Report 2022**

**<https://www.synopsys.com/blogs/software-security/poor-software-quality-costs-us/>**

# This is problematic, but there's something more...



**Synopsys Report 2022**

<https://www.synopsys.com/blogs/software-security/poor-software-quality-costs-us/>

**This is problematic, but there's something more...**

**In typical software there are**

**25 errors**

**per 1000 lines of code**

**Synopsys Report 2022**

**<https://www.synopsys.com/blogs/software-security/poor-software-quality-costs-us/>**

# Who am I?



# Who I am?

- Lecture of Software Dependability (M.Sc. in Computer Science)
- Lecture of Sustainable Software Engineering (M.Sc. in Computer Science)
- Lecturer of Software Engineering for Secure Cloud Systems (M.Sc. in IT Security and Cloud Technologies)
- Lecturer of Object Oriented Programming (B.Sc. in Computer Science)



# Who I am?



3 Full Professors, 2 Associate Professors, 1 Assistant Professor, 2 Research Associates, 12 Ph.D. Students

## Education

- Software Engineering (B.Sc.)
- Object Oriented Programming (B.Sc.)
- Programming and Data Structures (B.Sc.)
- Fundamentals of Artificial Intelligence (B.Sc.)
- Software Engineering Advanced Techniques (M.Sc.)
- Software Metrics, Analytics, and Quality (M.Sc.)
- Software Project Management (M.Sc.)
- **Software Dependability (M.Sc.)**
- Sustainable Software Engineering (M.Sc.)
- Software Engineering for A.I. (M.Sc.)
- Computer Science Education (M.Sc.)

## Research

- Software Maintenance and Evolution
- Software Testing
- Performance Testing
- Blockchain Software Engineering
- Quantum Software Engineering
- Requirement Engineering & Artificial Intelligence
- Social Software Engineering
- Software Project Management
- Diversity & Inclusion of Software Systems
- Fairness of Software Systems

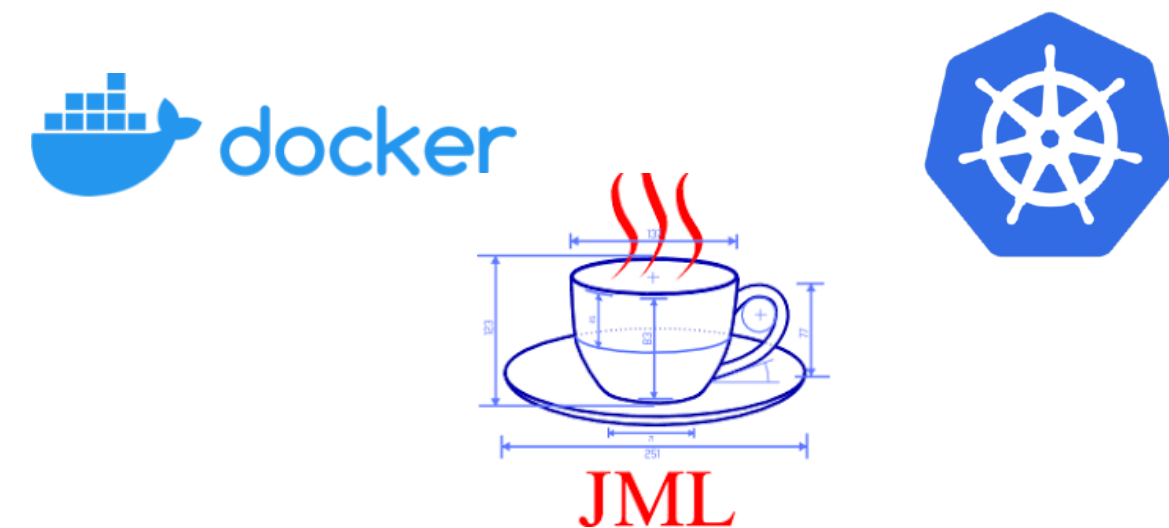


# The Need for Software Dependability

The dependability of a system reflects the extent of the user's confidence when operating and interacting with the system. A system can be considered dependable if (i) it operates without interruption, (ii) delivers the services that the stakeholders expect, (iii) does not adversely affect the environment, and (iv) does not damage its data or the system itself.

## M1: Fundamentals of Software Dependability

- Intro to Software Dependability
- Dependability Requirements, Faults, and Treatments
- Software Fault Avoidance in Specification and Implementation
- Software Fault Tolerance and Assessment
- IoT Architectures
- DevOps and Reliable Programmable Infrastructure



## M2: Software Testing

- Intro to Software Testing and Test Case Selection
- Test Case Assessment and Test-Driven Development
- Performance Testing
- Testing of Cyber-Physical Systems



## M3: Detection of Software Vulnerabilities

- Introduction to Software Vulnerabilities
- Dynamic Techniques to Detect and Exploit Software Vulnerabilities



# Administrative Information - General

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## lectures

Thu: 11:00 - 13:00  
Fri: 11:00 - 14:00

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## e-learning platform

<https://elearning.informatica.unisa.it/el-platform/course/view.php?id=1266>

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## email

[ddinucci@unisa.it](mailto:ddinucci@unisa.it)

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## office hours

Wed 14:00 - 16:00 and Fri 9:00 - 11:00  
Please contact me beforehand.  
Another option is to schedule a meeting via MS Teams.

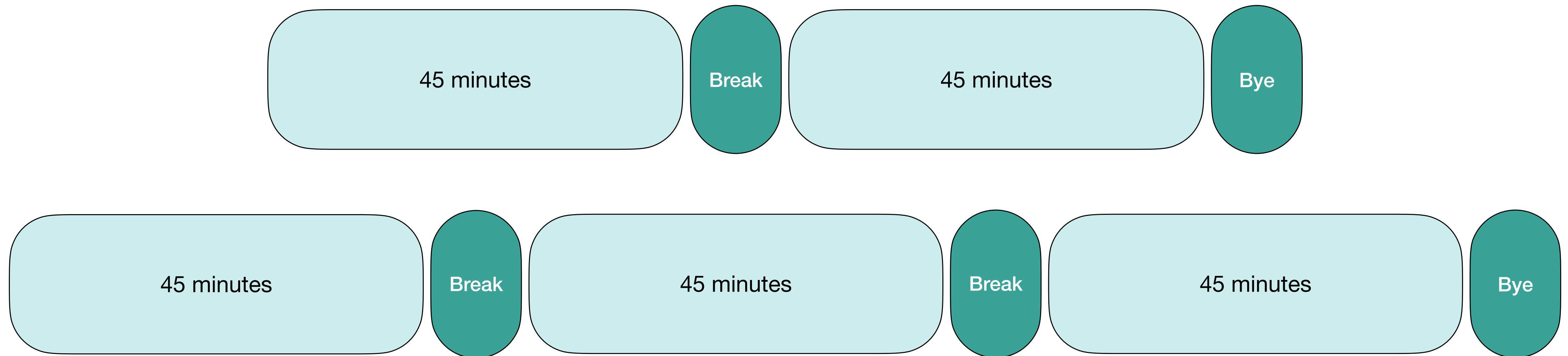
# Administrative Information - Calendar

|                                     | Thursday (11.00-13:00) F4                                         | Friday (11.00-14:00) F4                                                |
|-------------------------------------|-------------------------------------------------------------------|------------------------------------------------------------------------|
| <b>Week 1</b><br>(22 Sep - 26 Sep)  | Intro to the course                                               | Dependability Requirements, Faults, and Treatments                     |
| <b>Week 2</b><br>(29 Sep - 3 Oct)   | Software Fault Avoidance in Specification and Implementation      | JML and OpenJML                                                        |
| <b>Week 3</b><br>(6 Oct - 10 Oct)   | Software Fault Tolerance and Assessment                           | DevOps and Reliable Programmable Infrastructure                        |
| <b>Week 4</b><br>(13 Oct - 17 Oct)  | IoT Architectures                                                 | Docker, Docker Compose, and Podman                                     |
| <b>Week 5</b><br>(20 Oct - 24 Oct)  | Intro to Software Testing and Test Case Selection                 | Docker Swarm and Kubernetes<br>GitHub Workflows                        |
| <b>Week 6</b><br>(27 Oct - 31 Oct)  | Test Case Assessment and Test-Driven Development                  | JUnit and Mockito                                                      |
| <b>Week 7</b><br>(3 Nov - 7 Nov)    | Testing of Cyber-physical Systems                                 | JaCoCo, CodeCov, and PiTest                                            |
| <b>Week 8</b><br>(10 Nov - 14 Nov)  | Performance Testing                                               | Java Microbenchmark Harness, JU2JMH, AMBER                             |
| <b>Week 9</b><br>(17 Nov - 21 Nov)  | Introduction to Software Vulnerabilities                          | Code security and analysis in GitHub<br>Software Vulnerability Atelier |
| <b>Week 10</b><br>(24 Nov - 28 Nov) | Dynamic Techniques to Detect and Exploit Software Vulnerabilities | GitGuardian, Snyk, SonarQube Cloud                                     |
| <b>Week 11</b><br>(1 Dec - 5 Dec)   |                                                                   |                                                                        |
| <b>Week 12</b><br>(8 Dec - 12 Dec)  |                                                                   |                                                                        |
| <b>Week 13</b><br>(15 Dec - 19 Dec) |                                                                   | Flipped Classroom: Project Discussion                                  |

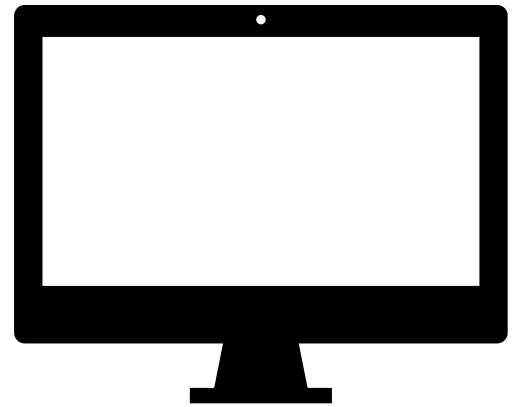
# Administrative Information - Lecture Structure

The ability to stay focused on contents that are conveyed to us and that require constant mental effort, focused on a goal, is known as *sustained attention*. To keep such a sustained attention constantly high, it is recommended to interleave 45 minutes of lecture with 15 minutes of break.

Hence, whenever possible, the lecture will be structure along those lines:

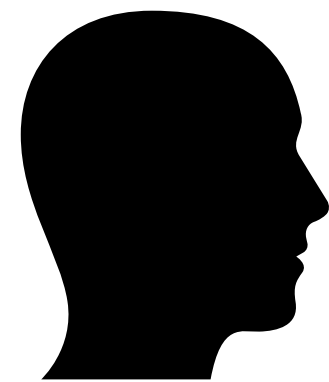


# Administrative Information - Exam Policy



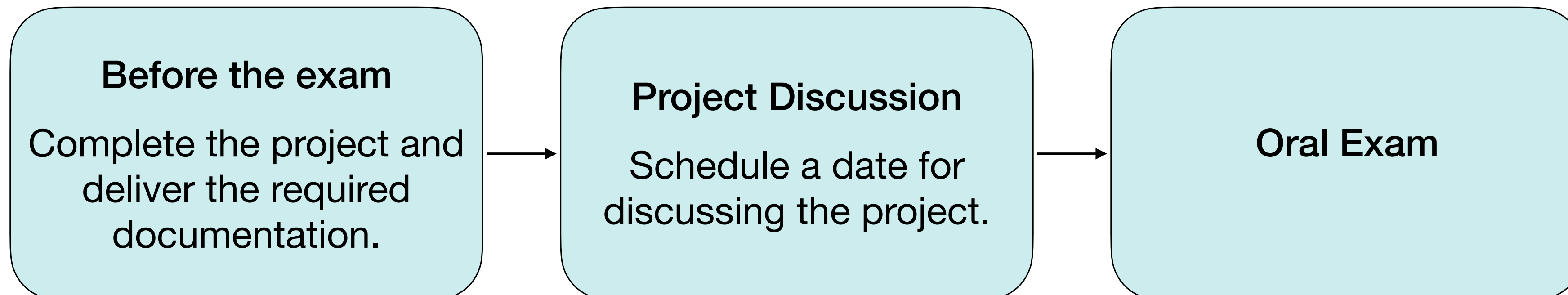
## Project

It will be the core of the exam. The grade will contribute to 50% of the final grade.  
You **must** present the project before the oral exam.

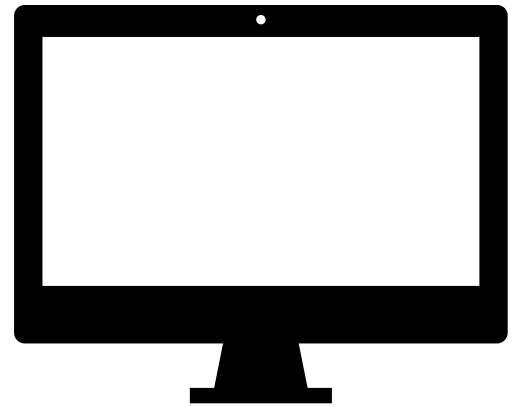


## Oral Exam

It will cover all aspects of the course and be composed of 3/4 questions.



# Administrative Information - Exam Policy



## Project

It will be the core of the exam. The grade will contribute to 50% of the final grade.  
You **must** present the project before the oral exam.

You have **three options**.

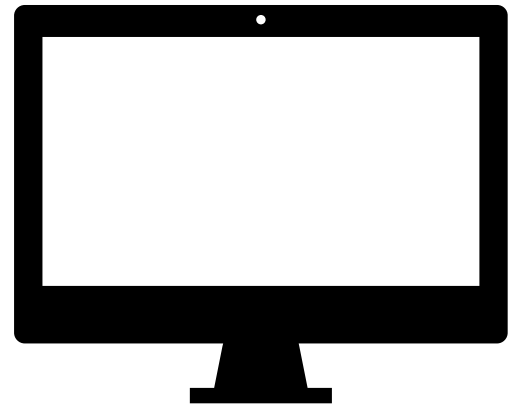
**Option #1 - Dependability analysis of a Java open-source project.** You will learn which projects are worth analyzing during the lab sessions. However, please double-check with me the suitability of the project BEFORE starting! Some open-source projects might be too trivial/complicated to touch/test —> be smart and ask first!

**Option #2 - Dependability analysis of your Java project.** If you developed a complex enough project, you could apply the methods studied in the course directly to that.

**Option #3 - Have a project in combination with other courses.** If you are developing a project involving software dependability concepts for other courses, let us discuss it.

The projects can be conducted in teams composed of **2/3 people or alone**.

# Administrative Information - Group Projects



## Project

It will be the core of the exam. The grade will contribute to 50% of the final grade.  
You **must** present the project before the oral exam.

### Option #1 and #2.

The goal is to measure the dependability of these systems through empirical investigations of the source code or the related change history, report on these aspects, and possibly provide developers with insights on how to structure the code or the development process better.

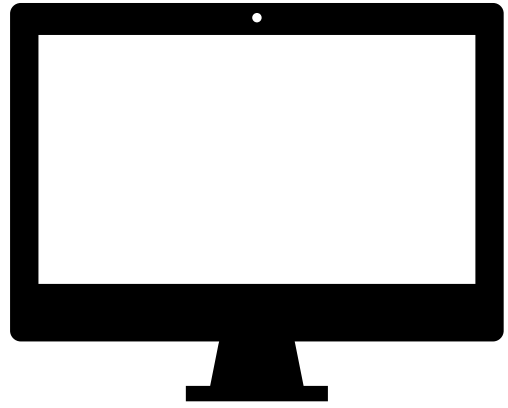
Depending on the projects, you may also require developing code that may be introduced in the codebase of the considered systems. In those cases, you may want to open pull requests and become open-source contributors.

Your report must cover the concepts of the course; in particular, you must apply the tools shown in the lab sessions to the selected software system. For each applied tool, you will receive some points that will compose the final project grade.

### Option #3.

The final grade will be independent of the other evaluations; particularly, it will focus on the attention to software dependability in the developed project.

# Administrative Information - Project Reporting



## Project

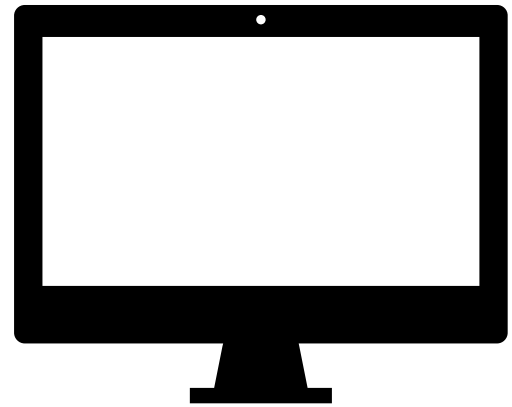
It will be the core of the exam. The grade will contribute to 50% of the final grade.  
You **must** present the project before the oral exam.

Each group must create a GitHub repository for all documents and code. If there is no repository at the end of the semester or the repository is incomplete, **this part of the exam will be considered failed.**

Each member of the group must contribute significantly to the project. We will analyze the git logs to verify your contribution: no commits means no contribution.

As such, **using Git appropriately is key.** Forget USB pens, Google Drive, or Dropbox folders!

# Administrative Information - Project Reporting



## Project

It will be the core of the exam. The grade will contribute to 50% of the final grade.  
You **must** present the project before the oral exam.

Each team should deliver a document reporting the project's methodology and execution.

Although this document has no fixed template, **you are strongly encouraged** to use LaTeX.

I strongly recommend using a template that can ease your activities. Specifically, I recommend the following template:

=> <https://www.acm.org/publications/proceedings-template>

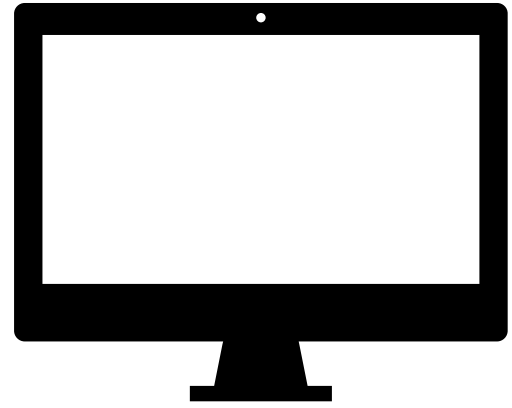
=> If so, please use, as format, the instructions:

```
\documentclass[sigconf]{acmart},  
\bibliographystyle{ACM-Reference-Format}.
```

The only reason to follow these recommendations is to help you focus on the actual content of the report rather than wasting time/effort dealing with references/formatting issues/others.

**The report must be stored on GitHub**, along with the other artifacts.

# Administrative Information - Project Presentation



## Project

It will be the core of the exam. The grade will contribute to 50% of the final grade.  
You **must** present the project before the oral exam.

At the end of the course, you need to plan for a final presentation before the oral examination. It is up to you to plan the final presentation, but this should be done **at least three days before the oral examination**.

The presentation must cover (i) the context of the project, (ii) the goals of the project, (iii) methodological steps conducted to address the goals, (iv) results and findings, and (v) conclusions.

**The presentation slides must be stored on GitHub**, along with the other artifacts.

# Administrative Information - What are my expectation?

Study lesson by lesson and apply theoretical concepts as they are introduced.  
Come to the lab sessions and get your project done step by step.

Do you have any doubt?

**Contact me via mail**

**Talk with me at the end of  
the lecture**

**Schedule a meeting with  
me (in presence or online)**



A close-up photograph of a white computer keyboard. The central focus is a large, rectangular white key with rounded corners. On this key, there is a dark blue icon of a coffee cup with three wavy lines above it representing steam. Below the icon, the word "Break" is printed in a dark blue, serif font. The key is set against a light-colored, textured keyboard surface. Other keys are visible in the background, including one with a double quote symbol and another with a four-way arrow symbol.

Break